

Notes: Altermatt, van Buggenum & Voellmy (JFE, 2024)
**Systemic bank runs without aggregate risk: How a misallocation of liquidity
may trigger a solvency crisis**

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1 Big picture

- **Question.** Can a systemic bank run arise without aggregate fundamental risk, and can it turn a liquidity problem into a solvency crisis?
- **Core mechanism.** Depositors use bank deposits as payment instruments. In a run, some patient depositors withdraw cash even though they do not need to consume. Cash is then misallocated away from impatient consumers who need it to buy goods.
- **Real-side feedback.** When impatient consumers cannot withdraw enough cash, aggregate demand in the decentralized goods market falls. Entrepreneurs sell less, some cannot repay bank loans, and bank assets become impaired.
- **Self-fulfilling logic.** The expected fall in bank loan repayments rationalizes patient depositors' decision to run. The run creates the solvency losses that make the run privately optimal.
- **Main novelty.** The return on bank assets is endogenous because bank borrowers' revenues depend on depositors' ability to withdraw and spend.
- **Deposit freezes.** A full freeze stops an ongoing run mechanically, but the prospect of a freeze may not prevent a run. Freezes reduce impatient consumers' cash access, which reduces demand and worsens loan losses.
- **Redemption penalties.** A partial freeze combined with a redemption penalty can stop a run ex post and sometimes prevent it ex ante, but the conditions are delicate.
- **Emergency liquidity.** Public liquidity provision can stop runs and may prevent them if it stabilizes loan repayments. Its effectiveness depends on whether asset purchases are inflationary and whether loan contracts are nominal or real.
- **Policy implication.** Preventing a run is different from stopping a run after it starts. Policies that block withdrawals can be self-defeating if they worsen the real demand channel that supports bank solvency.
- **Economic takeaway.** Bank fragility can come from a misallocation of means of payment, not only from aggregate asset risk or costly asset liquidation.

2 Environment and measurement objects

2.1 Agents and markets

- Time is divided into a centralized market (CM) and a decentralized market (DM).
- Households are ex ante identical but learn whether they are impatient or patient.
- Impatient households want to consume the DMgood and need liquid means of payment.
- Entrepreneurs borrow from banks, use capital and labor, and sell DMgoods to households.
- Banks take assets and issue demand deposits that households use to obtain cash for DMpurchases.

Interpretation. The model connects banks' liabilities and assets through the payment system: deposits fund households' spending, and that spending determines banks' loan repayments.

2.2 Deposit role in the payment system

The DMcash-in-advance condition is

$$p_t q_t \leq m_t, \tag{1}$$

where p_t is the DMgoods price, q_t is DMconsumption, and m_t is cash.

Interpretation. If impatient households fail to withdraw enough cash, they cannot buy goods. The demand shortage is not just a distributional issue; it directly affects bank borrowers' revenues.

2.3 Production and goods demand

Entrepreneurs produce the DMgood using capital and labor. The numerical examples use a Cobb-Douglas form:

$$f(k, h) = k^\alpha h^{1-\alpha}. \quad (2)$$

Active entrepreneurs choose supply q^e to maximize operating surplus:

$$\rho q^e - c(q^e, k_S), \quad (3)$$

where ρ is the real DMprice.

Interpretation. Entrepreneurs either produce and repay in full or do not produce and default. In a run, lower household spending lowers the return to producing and can trigger defaults.

2.4 Banking equilibrium types

The paper distinguishes three steady-state banking equilibria:

- **Full liquidity insurance (FLI):** $i = \iota$.
- **Partial liquidity insurance (PLI):** $i \in (0, \iota)$.
- **Zero lower bound (ZLB):** $i = 0$.

The steady state solves

$$\frac{u'(q)}{c_q(1, \kappa)} = 1 + \frac{\theta \iota (1 + i) + (1 - \theta)(\iota - i)}{\theta(1 + i)}, \quad (4)$$

$$-c_k(1, \kappa) = \frac{1 + i}{1 + \iota}. \quad (5)$$

Interpretation. FLI means banks can fully insure liquidity needs. ZLB and PLI equilibria have tighter asset scarcity or liquidity constraints, making the patient depositors' incentive constraint tighter.

2.5 Asset-market clearing

Let $\mathcal{A}$ denote real aggregate asset supply. Asset-market clearing implies

$$\mathcal{A} \begin{cases} \geq (1 - \theta)\rho q, & i = \iota \quad (\text{FLI}), \\ = \frac{1 + \iota}{1 + i}(1 - \theta)\rho q, & i \in (0, \iota) \quad (\text{PLI}), \\ \leq (1 + \iota)(1 - \theta)\rho q, & i = 0 \quad (\text{ZLB}). \end{cases} \quad (6)$$

Interpretation. Asset abundance supports FLI. Asset scarcity pushes the economy toward ZLB conditions, where fragility is especially sharp.

3 Models and theoretical specifications

3.1 Definition of a run

A run is a zero-probability event in which all depositors try to withdraw in the DM. Depositors arrive in random order, and banks serve them sequentially. Let:

- $\lambda \in [0, \theta]$ be the fraction of depositors served before banks or the government detect the run;
- d_S^I be the steady-state DMpayout to impatient depositors;
- M^{RE} be the total cash paid out to impatient depositors in a run;
- χ be the share of entrepreneurs that produce and repay.

Interpretation. The run reallocates cash from consumers with spending needs to patient depositors who hoard it until the next centralized market.

3.2 Run without a change in the money supply

If the amount of money in circulation does not change, then the real purchasing power of money remains unchanged:

$$\pi = \pi_S, \quad \rho = \rho_S. \quad (7)$$

The share of active entrepreneurs is

$$\chi = \frac{M^{RE}}{\theta d_S^I}. \quad (8)$$

Interpretation. With unchanged prices, real demand falls one-for-one with the cash reaching impatient consumers. Fewer active entrepreneurs means more loan defaults.

3.3 Run with changing real prices

For any real DMprice ρ , define the capital share of active entrepreneurs:

$$\alpha(\rho) = \frac{\rho q^e(\rho) - c(q^e(\rho), k_S)}{\rho q^e(\rho)}. \quad (9)$$

Then

$$\chi = \min \left\{ \frac{\alpha(\rho)}{\alpha(\rho_S)} \frac{M^{RE}}{\theta d_S^I}, 1 \right\}. \quad (10)$$

Interpretation. This formula isolates two forces: cash reaching impatient consumers and the distribution of revenues between labor compensation and loan repayment.

3.4 Fragility condition

The banking system is fragile if patient depositors get a strictly higher payoff by withdrawing in the DM than by waiting for the CM in a run.

- In ZLB and PLI equilibria, the system is fragile under pure deposit freezes for all detection thresholds and freezes.
- In FLI equilibria, banks may have a buffer, so fragility depends on aggregate asset supply and the composition of bank assets.

Interpretation. The most fragile cases are those in which patient depositors' steady-state incentive constraint already binds. Any losses on bank assets then make early withdrawal attractive.

3.5 Deposit freezes

Let η be the share of a deposit that may be redeemed in the DM after banks detect a run. Pure deposit freezes restrict withdrawals but do not charge an explicit redemption penalty.

The paper shows that pure freezes can stop a run mechanically, but they can fail to prevent runs from starting because they lower DM spending:

$$\text{freeze} \Rightarrow \text{less cash to impatient households} \Rightarrow \text{lower demand} \Rightarrow \text{loan losses} \Rightarrow \text{run incentives.} \quad (11)$$

Interpretation. This overturns the classic Diamond-Dybvig logic that suspension of convertibility eliminates belief-driven runs when there is no aggregate uncertainty.

3.6 Redemption penalties

Now let σ be a redemption penalty charged on DM redemptions. A policy (η, σ) stops a run if a patient depositor prefers not to withdraw after the restriction has been imposed:

$$\eta d_S^I + (1 - \eta - \sigma) d_R^P(\eta, \sigma) \leq d_R^P(\eta, \sigma). \quad (12)$$

If the run stops and remaining impatient depositors withdraw, then

$$M^{RE} = \theta[\lambda + (1 - \lambda)\eta] d_S^I, \quad (13)$$

and with unchanged real purchasing power,

$$\chi = \lambda + (1 - \lambda)\eta. \quad (14)$$

Interpretation. The penalty redistributes value away from DM redeemers toward those who wait. It can deter patient depositors, but if it is too high, impatient depositors may not withdraw enough to support goods demand.

3.7 Stopping versus preventing runs

The paper's policy distinction is:

- **Stopping** a run means ending it once the policy is imposed.
- **Preventing** a run means removing patient depositors' ex ante incentive to join the run before the policy is imposed.

Proposition 6 says there always exists a deposit-freeze/redemption-penalty policy that stops a run. Proposition 7 says there may be no such policy that prevents a run, even if the run is detected immediately.

Interpretation. Policies that stop runs ex post are not automatically credible deterrents ex ante because depositors care about the real losses created before and after restrictions.

3.8 Emergency liquidity

The government can provide liquidity by buying bonds and loans from banks. The discounts on bond and loan purchases are δ_b and δ_ℓ , with $\delta_b \geq \delta_\ell$.

The redemption rule induced by emergency liquidity is

$$\eta = 1 - \sigma = \min \left\{ \frac{m_S^b - \theta d_S^I + \delta_b(1 + i_S)b_S^b + \delta_\ell(1 + i_S)\ell_S^b}{(1 - \theta)d_S^I}, 1 \right\}. \quad (15)$$

The cash reaching impatient depositors in a run is

$$M^{RE} = \theta[\theta + \eta(1 - \theta)]d_S^I. \quad (16)$$

Interpretation. Emergency liquidity supports banks' payout capacity, but if financed by new money it can be inflationary. Inflation can reduce real purchasing power and therefore still induce defaults.

3.9 Emergency-liquidity results

- **Non-inflationary liquidity.** If loan purchases are sufficiently discounted, emergency liquidity is not inflationary and stops runs.
- **Prevention in ZLB/PLI.** Non-inflationary emergency liquidity prevents runs in ZLB and PLI equilibria only if the government is as efficient as banks at collecting loan repayments.
- **Inflationary liquidity.** Face-value purchases can prevent runs if inflation does not reduce the capital share $\alpha(\rho)$.
- **Real loan contracts.** If loans are denominated in real terms, face-value emergency liquidity prevents runs.

Interpretation. The lender of last resort is powerful only if the intervention stabilizes the real cash flow backing bank assets, not merely the nominal withdrawal capacity of banks.

4 Identification and theoretical design

4.1 What makes the mechanism internally credible

- The model closes the loop between banks' liabilities, goods demand, entrepreneurial revenues, and bank asset payoffs.
- There is no aggregate fundamental risk. Runs are self-fulfilling because the run itself creates the loan losses.
- Sequential service and random arrival make early withdrawal privately valuable once losses are expected.
- The decentralized goods market gives cash a real payment role, so liquidity misallocation has output consequences.

Interpretation. The central innovation is not another asset-liquidation loss. It is a payment-system externality: the wrong depositors get the liquidity.

4.2 Why deposit freezes are self-defeating

- A freeze reduces the number of redemptions.
- But some of the constrained redemptions are by impatient depositors.
- Their lower cash balances reduce DMdemand.
- Lower demand reduces loan repayment.
- The expected loan losses restore patient depositors' incentive to run before the freeze.

Interpretation. Freezing a liability can protect a bank only if the bank's assets are independent of the frozen payments. In this model, they are not.

4.3 Why redemption penalties can work

- A redemption penalty discourages patient depositors from continuing to run.
- It redistributes resources to depositors who wait.
- If set carefully, it still allows impatient households to spend enough to prevent excessive defaults.
- If set too high, it deters useful spending and can be counterproductive.

Interpretation. The instrument has to balance two constraints: deter patient withdrawals and preserve impatient consumption.

4.4 Why emergency liquidity can work

- Emergency liquidity lets banks serve withdrawals without selling assets too deeply or cutting off spending.
- If it keeps nominal and real purchasing power stable, entrepreneurs repay and the run is not rational.
- If it is inflationary and reduces real purchasing power, the policy may still fail unless nominal debt burdens fall in a stabilizing way.

Interpretation. Liquidity provision must stabilize the asset side of bank balance sheets through the real economy.

4.5 What the paper cannot fully prove

- The paper is theoretical; it does not estimate the mechanism structurally.
- The model is most direct for economies where payments require cash or cash-like redemption of deposits.
- The system is treated as a consolidated banking sector reacting jointly once a run is detected.
- The run is a zero-probability off-equilibrium event, so policy credibility and implementation delays are stylized.
- The empirical discussion is suggestive rather than a direct test.

Interpretation. The contribution is a mechanism and policy logic, not an empirical measurement exercise.

5 Tables and figures

5.1 Figure 1 and Table 1: an example where runs can be prevented

Read:

- The horizontal axis is η , the share of deposits that can be redeemed after restrictions are imposed.
- The vertical axis is σ , the redemption penalty.
- The green dotted curve is the minimum penalty that stops a run.
- The red dashed curve is the minimum penalty that prevents a run.
- The blue curve is the maximum penalty impatient depositors are willing to pay.
- The grey region is the set of policies that both stop and prevent runs.
- Table 1 reports the parameter values used for this example: the economy is in a ZLB equilibrium.

Economic message. A suitable mix of partial convertibility and redemption penalties can prevent a run only if the lower-bound and upper-bound constraints overlap.

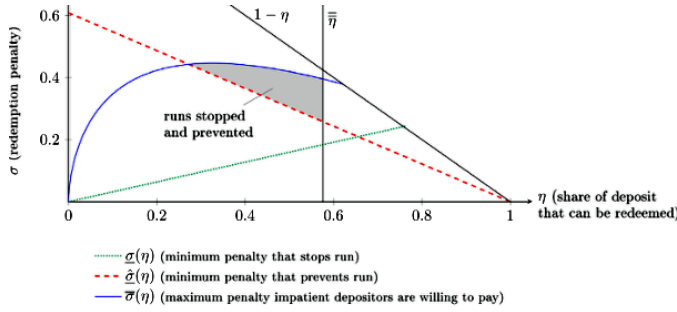


Fig. 1. Example where runs can be prevented.

(a) Figure 1: policies that stop and prevent a run.

parameter values for Fig. 1.

α	ν	θ	π	λ	t	B	q_S	ρ_S	κ_S
0.6	0.65	0.6	0.6	0.45	0.05	0	0.037	1.904	1.199

(b) Table 1: parameter values.

5.2 Figure 2 and Table 2: an example where runs cannot be prevented

Read:

- Figure 2 uses a similar policy diagram but has no grey region.
- The “run stopped” region exists, but it does not overlap with the region that prevents runs ex ante.
- The lack of overlap occurs even when banks can identify the run immediately in the numerical example.
- Table 2 reports the parameter values for the second numerical example.

Economic message. Stopping an ongoing run does not guarantee that the threat of the intervention prevents a run from starting. The demand-loss channel can make prevention impossible.

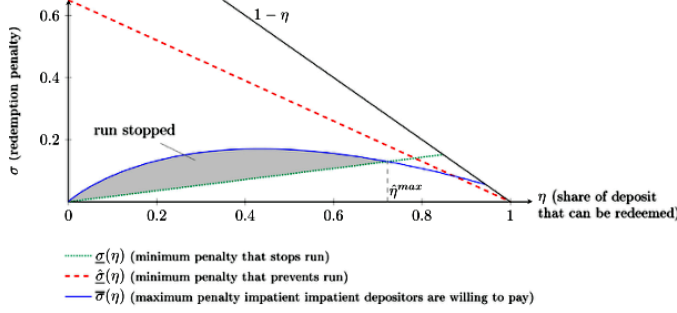


Fig. 2. Example where runs cannot be prevented.

(a) Figure 2: no policy simultaneously stops and prevents the run.

Parameter values for Fig. 2.

α	ν	θ	n	λ	τ	B	q_S	ρ_S	κ_S
0.65	0.95	0.6	0.6	0	0.02	0	5.725×10^{-7}	1.886	1.251

(b) Table 2: parameter values.

6 Short summary

- The paper builds a general equilibrium model of systemic bank runs without aggregate fundamental risk.
- Banks lend to entrepreneurs, entrepreneurs sell goods to households, and households pay by redeeming deposits.
- This creates a tight link between deposit withdrawals and bank loan repayment.
- In a run, some cash goes to patient depositors who do not spend it.
- Impatient depositors have less cash, demand falls, entrepreneurs default, and banks incur losses.
- These losses rationalize the patient depositors' initial decision to run.
- Pure deposit freezes do not necessarily prevent runs because they can reduce aggregate demand and worsen loan repayment.
- Redemption penalties can stop runs and sometimes prevent them, but the policy must deter patient depositors while preserving impatient consumption.
- Emergency liquidity can be effective if it stabilizes loan repayments and real purchasing power.
- Non-inflationary liquidity always stops runs, but prevention is harder in ZLB/PLI equilibria unless public asset purchases are efficient.
- Face-value liquidity can prevent runs if inflation does not reduce the capital share; with real loan contracts, it prevents runs.
- The mechanism is most relevant for cash-intensive economies or settings where deposit convertibility is necessary for payments.

7 Conclusion

7.1 Core conclusion

Altermatt, van Buggenum, and Voellmy show that a systemic run can be self-fulfilling even without aggregate risk. The run misallocates liquidity across households, reduces goods demand, causes borrowers to default, and thereby generates the bank losses that make running optimal.

7.2 Economic intuition

The intuition is that banks' assets depend on the functioning of their liabilities. Deposits are not merely funding instruments; they are also payment instruments. If a run prevents the right depositors from using

deposits to buy goods, bank borrowers lose revenue and cannot repay. A liquidity misallocation therefore becomes a solvency problem.

7.3 Takeaway

The paper changes how one should think about run prevention. Policies that only restrict withdrawals may be self-defeating if they damage the real demand that supports bank assets. Effective intervention must preserve the payment function of deposits and stabilize borrowers' repayment capacity.